ELTAC401 – HEATING, VENTILATION, AND AIR CONDITIONING (HVAC) SYSTEM INSTALLATION

DATE: .../.../2025

PERIOD: .....

TECHNICAL SECONDARY SCHOOLS (TSS)

END OF TERM I EXAMINATIONS: 2025-2026

QUESTION PAPER

TRADE NAME: ELT&ETE

RQF LEVEL: 4

MODULE TITLE: **HEATING, VENTILATION, AND AIR CONDITIONING (HVAC) SYSTEM INSTALLATION**

MARKS: **...../100**

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed Sections (A, B, C). Follow the instructions given below, and answer the indicated questions in each section for a total of 100marks

Section **A**: 26 questions, all are **Compulsory**. **55 marks**
Section **B**: Five 5 questions, choose and answer any three (3). **30 marks**
Section **C**: Two 2 questions, choose and answer any one (1) **15 marks**

Allowed Materials: Blue pen, Ruler, Calculator, Pencil, (Fitting with the exam paper)

Section **A**: 26 questions, all are **Compulsory**. **55 marks**

Choose the correct answer. ***(20 MARKS)***

1. HVAC stands for:
   a) Heat, Voltage, Air Control
   b) Heating, Ventilation, and Air Conditioning
   c) Heat Variable Air Compressor
   d) Humidity, Velocity, Air Circulation
2. The component responsible for mixing outdoor air and return air in an AHU is the:
   a) Preheat coil
   b) Mixing section
   c) Cooling coil
   d) Bag filter
3. The primary function of a cooling coil is to:
   a) Remove dust
   b) Add heat to the air
   c) Lower the temperature of air
   d) Increase humidity
4. AFUE is a measure of:
   a) Air flow rate
   b) Furnace efficiency
   c) Humidity percentage
   d) Ventilation capacity
5. NFPA 70 refers to:
   a) Refrigerant classification
   b) Electrical safety requirements
   c) Fire damper installation
   d) Indoor air quality standards
6. Which of the following is a combustion safety risk?
   a) Low airflow
   b) Gas leaks
   c) Dirty filters
   d) Poor insulation
7. Refrigerant toxicity and flammability classification is defined by:
   a) ASHRAE 62
   b) ISO 817:2014
   c) ASHI guidelines
   d) NFPA 90A
8. The heating component of HVAC typically uses:
   a) Evaporator
   b) Furnace or boiler
   c) Reheat coil only
   d) Ventilation duct
9. A central HVAC system serves:
   a) Only one room

b) One or more thermal zones
c) Only the ventilation system
d) Only cooling functions

10. Which of the following is a type of refrigeration system?
    a) Direct expansion
    b) Combustion circulation
    c) Heating conduction
    d) Forced ventilation

Write TRUE or FALSE. ***(20 MARKS)***

11. AHU is a component of the HVAC system.
12. Cooling coils increase the temperature of air before supply.
13. Ventilation removes stale air and brings fresh air into a building.
14. OSHA standards apply only to electrical safety.
15. Combustion appliances can produce carbon monoxide.
16. Absorption refrigeration uses mechanical compression.
17. Refrigerants can pose toxicity and flammability hazards.
18. Smoke alarms should be installed in kitchen areas for safety.
19. A low gas-pressure switch ensures safe ignition of the burner.
20. Central HVAC systems require multiple thermostats.

FILL IN THE BLANKS ***(10 MARKS)***

21. The efficiency of a furnace is measured using _____.
22. The coil used to warm fresh outdoor air before conditioning is the _____ coil.
23. HVAC refrigerant safety classification is defined under _____ 817:2014.
24. The four basic refrigeration cycle components are compressor, condenser, evaporator, and _____.
25. Ventilation controls indoor air quality by removing moisture and _____.

MATCHING ***(5G MARKS)***

Match Column A to Column B.

Column A

A. Cooling coil
B. Bag filter
C. ASHRAE

D. Expansion valve
E. Smoke damper

Column B

1. Removes larger airborne particles
2. Prevents spread of fire through ducts
3. Regulates refrigerant flow
4. Industry standards for HVAC
5. Lowers air temperature

Section **B**: Five 5 questions, choose and answer any three (3). **30 marks**

Provide clear, complete answers.

1. Explain the role and importance of HVAC systems in indoor comfort and air quality.
2. Describe the main components of an AHU and state their functions.
3. Explain the working cycle of a basic HVAC cooling system.
4. Discuss three types of combustion safety problems and how they affect occupants.
5. State the major guidelines (ASHRAE, NFPA, ASHI) related to HVAC safety and their importance.

Section **C**: Two 2 questions, choose and answer any one (1) **15 marks**

1. Compare vapour compression and absorption refrigeration systems.
2. Differentiate between central and local HVAC systems with examples.